- Connect Laravel containers to a MySQL database running on the host machine
- Run the Laravel 12 application in a production container setup

17.8.3 Prerequisites

- Laravel 12 project (can be pushed to GitHub or zipped)
- Ubuntu Server 22.04+ with root or sudo access
- MySQL installed and running on the host machine (e.g., `localhost:3306`)
- Database created for Laravel (e.g., `laravel12_db`), with user and password
- Port `80` open and accessible

17.8.4 Steps

Here's a step-by-step guide to deploying Laravel 12 in Docker on Ubuntu Server.

17.8.4.1 Step 1: Install Docker and Docker Compose (Latest Version)

This installation method based on the official Docker documentation is the recommended way to install Docker Engine on Ubuntu. It ensures you get the latest version of Docker and its dependencies. Ref: https://docs.docker.com/engine/install/ubuntu/.

Before you install Docker Engine for the first time on a new host machine, you need to set up the Docker apt repository. Afterward, you can install and update Docker from the repository.

```
# Add Docker's official GPG key:
sudo apt-get update
sudo apt-get install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.as
sudo chmod a+r /etc/apt/keyrings/docker.asc


# Add the repository to Apt sources:
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://d
  $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update
```

To install the latest version, run:

```
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-comp
```